

Figure 1 Fence Specification Soft Ground

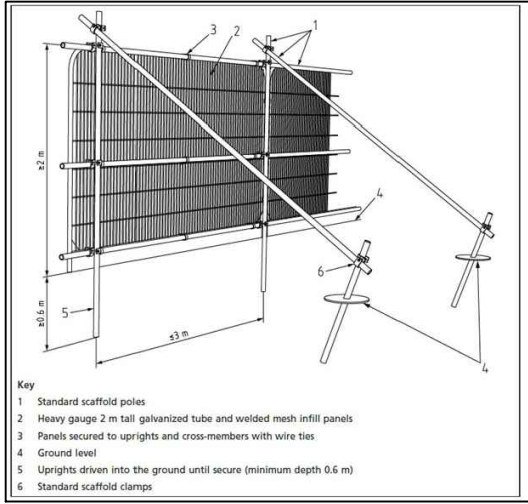
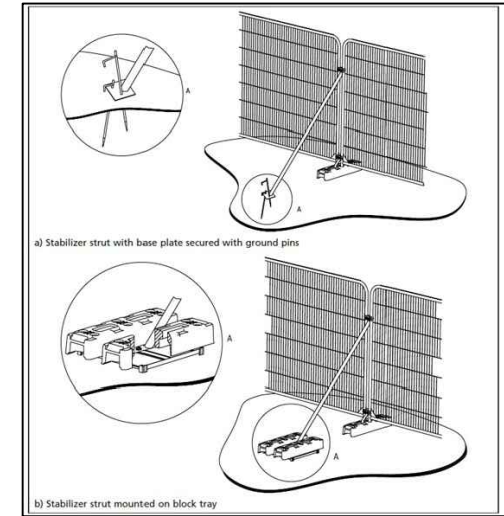


Figure 2 Fence Specification Hard Ground ONLY



2 Ground Protection Measures

2.1 The ground protection measures will be for pedestrian work access only. This will consist of a single thickness of scaffold boards placed either on top of a driven scaffold frame to form a suspended walkway, or on top of a compression-resistant layer (e.g., 100mm minimum depth of woodchip), laid onto a geotextile membrane.

2.2 Alternatively, Ground Guards or a similarly tested product, as detailed in Appendix VI could be used. This is accordance with BS 5837 (2012) and is to prevent compaction to the underlying soil.

2.3 If scaffolding is being installed then a false floor can be created just above ground level. This will act as ground protection preventing the soil underneath from being compacted by site activity, as shown in Figure 3 of Appendix I.

2.4 If a piling rig is required then the ground protection measures will be for pedestrian operated plant up to a gross weight of two tonnes. This will consist of proprietary interlinked ground protection boards placed on top of a compression-resistant layer (e.g., 150mm of woodchip) laid on a geotextile membrane. Alternatively Piling Mats can be used.

9 Hard Surfaces within the RPA

9.1 The car parking area within the RPA shall be constructed without soil compaction or soil stripping and laid in accordance with this Method Statement. A product such as GreenFix's GoeWeb or Geosynthetics Cellweb, or an alternative with evidence of its effectiveness at protection roots, shall be used. It shall be installed in full accordance with the manufacturer's specification.

9.2 The no-dig construction shall be undertaken in accordance with the manufacturer's specification and method statements.

9.3 Ground Preparation:

- All ground vegetation will be killed using a suitable herbicide to the required level, under the supervision of the project arboriculturalist.
- All dead organic material will be removed.
- All major protrusions will be removed. Stumps will be ground out.
- Fill major hollows with no fines 4/20mm clean angular stone.
- Place Geotextile over the area to be protected ensuring overlaps with a minimum of 300mm.
- Mark out areas to be protected with edging detail e.g., timber boards.

9.4 Installation Process:

- Lay GeoWeb (or equivalent i.e., Cellweb) over entire area of car parking area within the RPA, to extend 100mm beyond area width (see manufacturers specification), and pin with four metal pins along the width of the panel.
- Expand the panel over the geotextile extending to the required length, then pin across the opposite panel side.
- Pin along the length of the panel on all sides.
- If full panels are not being used, then ensure the cells have been expanded to their full dimensions.
- Staple or cable tie any adjacent panels together.
- The geocell panels can be cut to shape if required with a heavy-duty Stanley knife.

9.5 Filling the Geocell.

- Use 4/20mm or 40/20mm angular stone depending on the cell depth being used.
- Fill the cells with clean angular stone.
- Allow 25mm overfill for any settlement of the stone in the cells.
- If the area is to be trafficked immediately, slightly increase the amount of surcharge overfill to a maximum 50mm.
- This will be tipped from one end so that machinery moves on already spread sub-base and not upon the geogrid or ground close to the geogrid.
- Compact the sub-base using handheld vibrating tamper.

9.6 Apply Surface Dressing

There are various surface dressings that can be applied, and the manufactures guidance on how to apply each should be followed from the specification. Surface dressing include.

- Block paving
- Porous and standard asphalt
- Resin Bonded Gravels
- Loose gravel

1 Construction Exclusion Zone

1.1 No works will be undertaken within any Construction Exclusion Zone (CEZ). The CEZs are to be afforded protection at all times and will be protected by fencing. A protective fence shall be erected prior to the commencement of any site works e.g., before any materials or machinery are brought on site, development or the stripping of soil commences.

1.2 The fence shall have signs attached to it stating that this is a Construction Exclusion Zone and that **NO WORKS are Permitted within the fence**. The tree protection fencing may only be removed following completion of all construction works.

1.3 The fence is required to be sited in accordance with the Tree Protection Plan.

1.4 They must be constructed as per Figures 1 and 2 in BS 5837 2012 and be fit for excluding any construction activity. Any other fence or barrier used must be fit for the purpose.

1.5 The fencing unless otherwise agreed with the tree officer shall consist of Heras fencing panels, around 3.5m long and 2 m tall. They shall be fixed into the ground on scaffold poles driven at least 0.6 m into the ground. They shall be supported by rear struts also secured to posts driven into the ground.

1.6 All bolts shall be secured from inside the fencing to prevent easy removal from the outside during the construction phase.

1.7 Where there are **existing hard surfaces**, then rubber feet can be used to support the fencing, but these rubber feet shall be secured into the ground with road pins or other robust metal pins, to prevent the fencing being moved. This stall also be secured by rear struts which are also pinned into the ground.

1.8 All tree protection fencing shall be regarded as sacrosanct and will not be removed or altered without prior written consent of the Local Authority Tree Officer.

Figure 3 Example of Signage on Fencing



11 Foundation Designs

11.1 As there is construction in close proximity to some significant retained trees some form of piled, or cantilever foundation will be required to minimise the root disturbance. This could involve the use of mini piles or screw piles.

11.2 Due to the extensive RPAs within the proposed footprint of the building and extensive canopy overhead there shall be restrictions on the type of equipment that will be used. It will have to be a lightweight piling rig with a weight of less than 2 tonnes and it will be on rubber tracks. It shall have a rig height of no more than 8 m. The tree canopy clearance is greater than 8 m.

11.3 The proposed pile locations will have pilot holes excavated to ensure that there are no significant roots (>25mm in diameter) at these locations. This will be achieved using an air spade or hand tools.

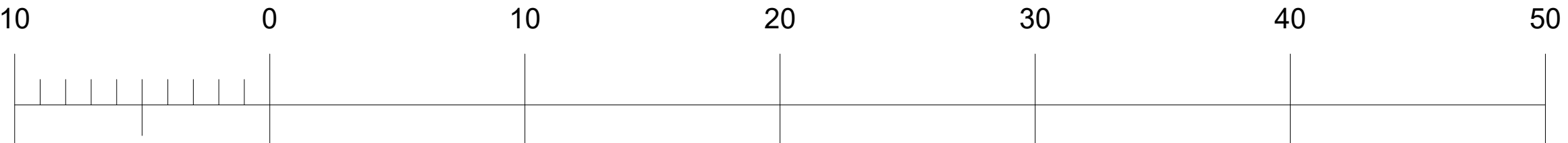
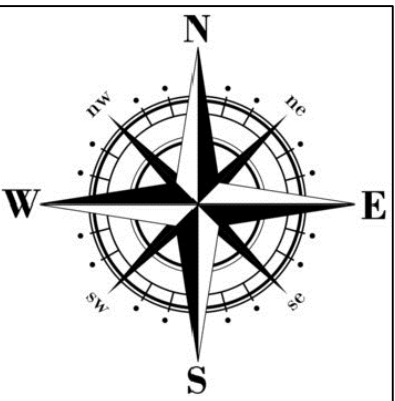
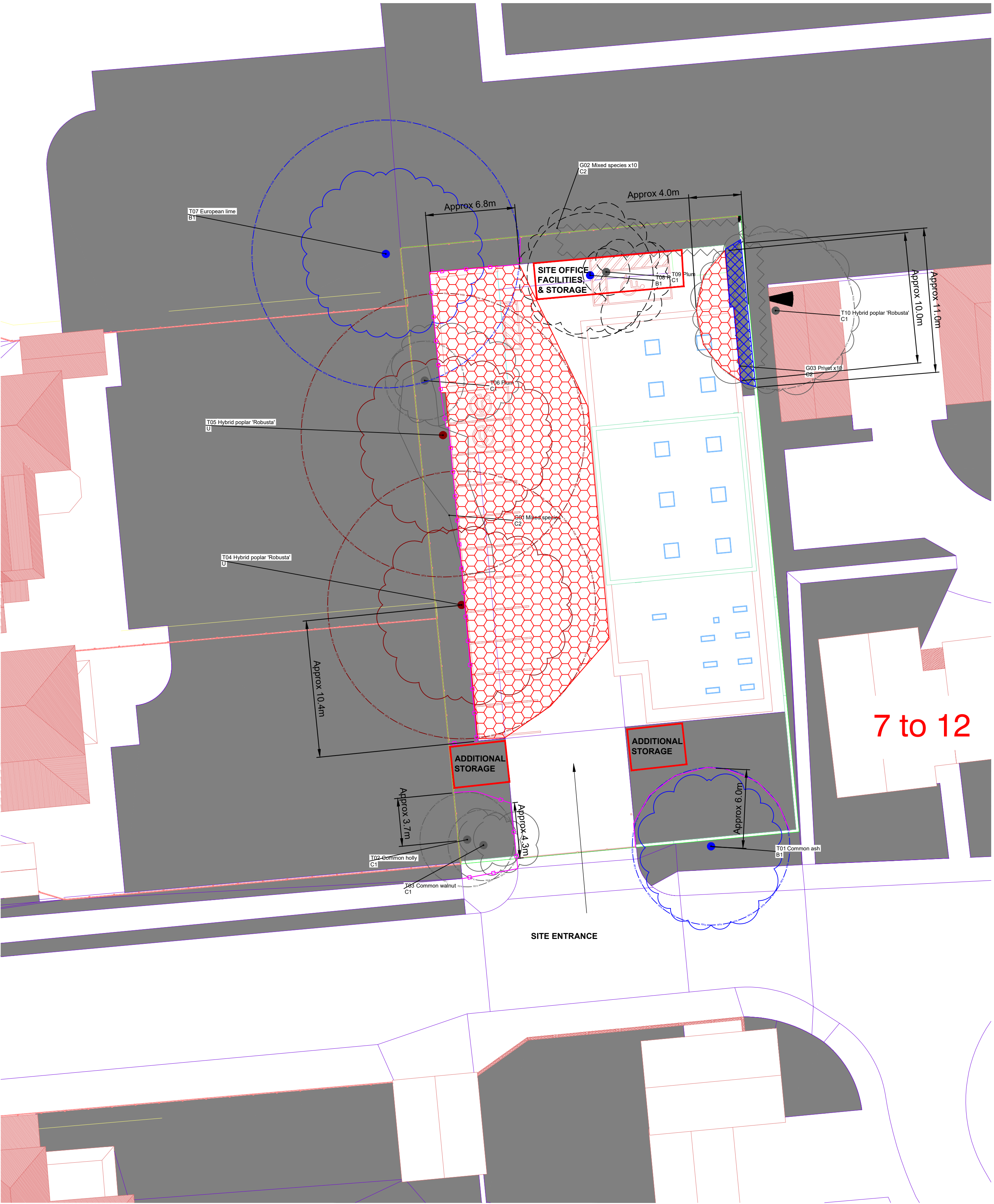
11.4 If there are significant structural roots encountered greater than 25mm in diameter then alternative pile locations will have to be identified and pilot holes excavated to ensure no significant tree roots are there.

11.5 The raft or beams may be located on the pile caps and all these elements will be above the existing ground level to prevent any further damage to tree roots. The only part that will be below ground is the piles themselves.

11.6 An impermeable layer will be placed underneath any raft or beams that are poured on site. This is to prevent leaching from the cement whilst it sets. If pre-cast rafts or beams are used, then this is not required.

11.7 Slabs for larger structures (e.g., dwellings) should be constructed with a ventilated air space between the underside of the slab and the existing soil surface (to enable gas exchange and venting through the soil surface).

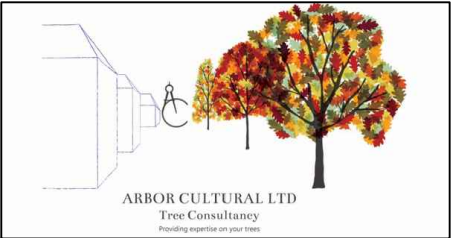
11.8 Specialist input on foundation design and the depth of foundations, pile numbers and locations will be required from a structural engineer, and they will have to be consulted if any pile locations are moved to avoid significant tree roots.



Legend

- CATEGORY A TREE
- CATEGORY B TREE
- CATEGORY C TREE
- CATEGORY U TREE
- ROOT PROTECTION AREA (RPA) Coloured by Tree Category
- MODIFIED ROOT PROTECTION AREA
- CROWN SPREAD
- TREE TO BE REMOVED
- PROTECTIVE FENCING
- TEMPORARY GROUND PROTECTION
- "NO-DIG" CONSTRUCTION
- NEW TREE
- SOIL TEST
- SHADE (CURRENT)
- SHADE (FUTURE)
- LOW BRANCH DIRECTION
- HAND- DIG EXCAVATION
- CONSTRUCTION EXCLUSION ZONE
- PROPOSED LAYOUT

Notes:



Arbor Cultural Ltd.
Providing Expertise on Your Trees (R)

36 Central Avenue, West Molesey, Surrey, KT8 2QZ
T 0333 577 5523 M 07899 984162
E: admin@arbor-cultural.co.uk W: www.arbor-cultural.co.uk

Client: Cheam Muslim Welfare Services Limited

Project: Former Ambulance Station, Dorset Road, Belmont, Sutton, SM2 6JH

Title: TREE CONSTRAINTS PLAN
TREE PROTECTION PLAN

Date: 15/12/25 Scale: 1: 200 Original Paper Size: A1

Drawn: IST Checked: - N/A Job Ref: AC.2025.861

Drawing Number: TPP-01 Rev: A

